

MOVIE ASSIGNMENT

1. Use movies11.csv,movie12.csv,movie13.csv and rating11.csv file and solve following

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Question 1

col_names = ['movieId', 'title', 'genres']

df1 = pd.read_csv('/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies11.csv')

df2 = pd.read_csv(
    '/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies12.csv',
    header=None,
    names=col_names
)
df3 = pd.read_csv(
    '/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies13.csv',
    header=None,
    names=col_names
)

rdf = pd.read_csv('/home/pgcp-esd/Downloads/PYTHON/movie assignment/rating11.csv')

print(df1)
print(df2)
print(df3)
print(rdf)

mdf = pd.concat([df1, df2, df3], ignore_index=True)

print(mdf)
g = rdf.groupby("movieId")
print(g)
print(g[['rating']].mean()) #to calculate columnwise mean
m=g[['rating']].mean()
print(m)
fdf=pd.merge(mdf,m,on="movieId",how="left")
```

2. find all masala movie - (action ,romance,comedy,thriller)

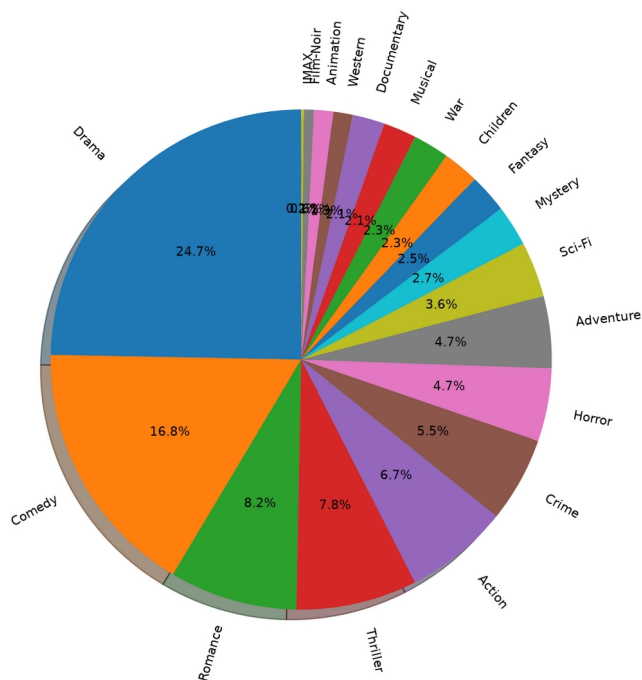
```
masala_mdf=mdf[mdf['genres'].str.contains('Action',case=False,na=False) &
mdf['genres'].str.contains('Romance',case=False,na=False) &
mdf['genres'].str.contains('Comedy',case=False,na=False) &
mdf['genres'].str.contains('Thriller',case=False,na=False)]
```

```
print(masala_mdf.info())
```

3. plot a pie chart to represent genre and frequency of movie count

```
genre_counts = mdf["genres"].str.split("|").explode().value_counts()
print(genre_counts)
```

```
plt.figure(figsize=(9, 9))
plt.pie(
    genre_counts,
    labels=genre_counts.index, # Genre names
    autopct='%1.1f%%',        # Automatically shows the percentage on each slice
    shadow=True,
    startangle=90,
    rotatelabels=True
)
plt.show()
```



4. find average rating for each movie then merge 2 frames, display movieid,name,rating

```
print(mdf)
g = rdf.groupby("movieId")
print(g)
print(g[["rating"]].mean()) #to calculate columnwise mean
m=g[["rating"]].mean()
print(m)
fdf=pd.merge(mdf,m,on="movieId",how="inner")
```

```
print(fdf)
```

5. draw pie chart for each genre and average rating

```
genre_avg_rating = (  
    fdf.assign(genres=fdf["genres"].str.split("|"))  
    .explode("genres")  
    .groupby("genres")["rating"].mean().round(2))  
  
print(genre_avg_rating)  
  
plt.figure(figsize=(9, 9))  
plt.pie(  
    genre_avg_rating,  
    labels=genre_avg_rating.index, # Genre names  
    autopct=lambda val: f"{val:.2f}", # Automatically shows the percentage on each slice  
    shadow=True,  
    startangle=90,  
    rotatelabels=True  
)  
  
plt.title("Average Rating by Genre", fontsize=14)  
plt.show()  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

Question 1

```
col_names = ['movieId', 'title', 'genres']
```

```
df1 = pd.read_csv('/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies11.csv')
```

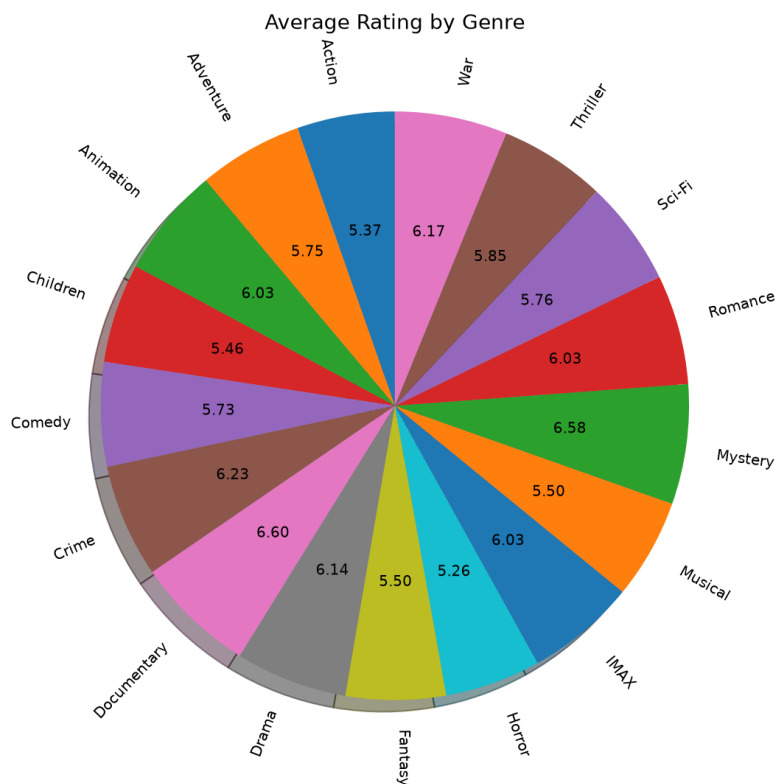
```
df2 = pd.read_csv(  
'/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies12.csv',  
header=None,  
names=col_names  
)  
df3 = pd.read_csv(  
'/home/pgcp-esd/Downloads/PYTHON/movie assignment/movies13.csv',  
header=None,  
names=col_names  
)
```

```
rdf = pd.read_csv('/home/pgcp-esd/Downloads/PYTHON/movie assignment/rating11.csv')
```

```
print(df1)
print(df2)
print(df3)
print(rdf)
```

```
mdf = pd.concat([df1, df2, df3], ignore_index=True)
```

```
print(mdf)
g = rdf.groupby("movieId")
print(g)
print(g[['rating']].mean()) #to calculate columnwise mean
m=g[['rating']].mean()
print(m)
fdf=pd.merge(mdf,m,on="movieId",how="left")
```



6. draw bar graph for each rating and number of movies

```

import plotly.express as px
import plotly.io as pio

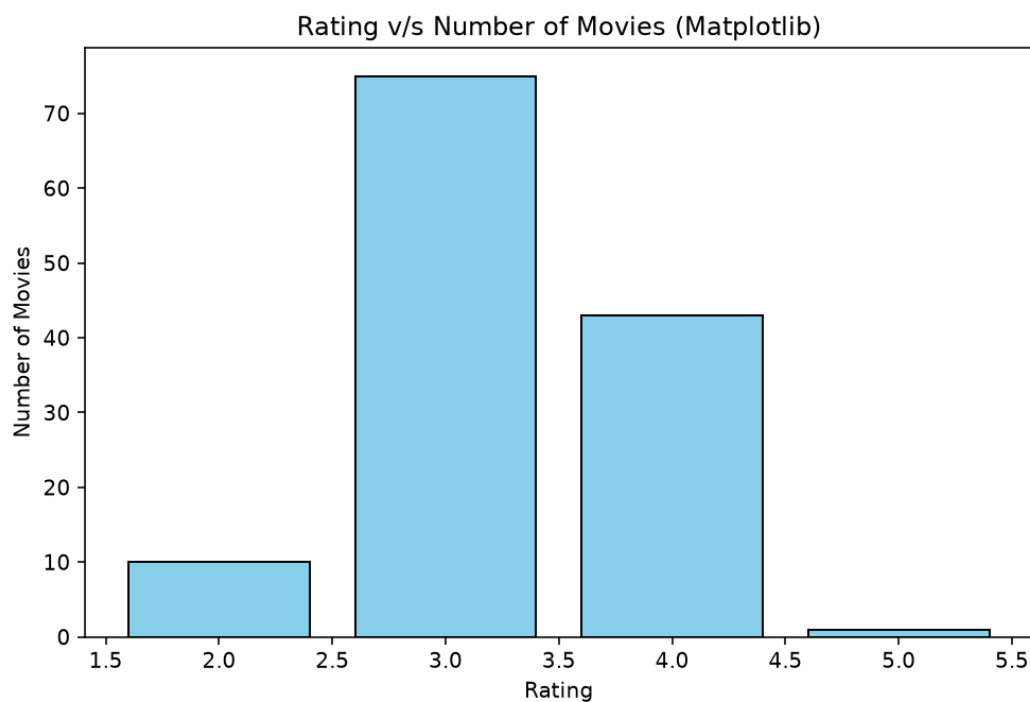
pio.renderers.default = "browser"

rating_count = fdf["rating"].round().value_counts().reset_index()
rating_count.columns = ['rating', 'count'] # Name the columns

plt.figure(figsize=(8, 5))
plt.bar(rating_count['rating'], rating_count['count'], color='skyblue', edgecolor='black')
plt.xlabel('Rating')
plt.ylabel('Number of Movies')
plt.title('Rating v/s Number of Movies (Matplotlib)')
plt.show()

fig = px.bar(
    rating_count,
    x='rating',
    y='count',
    title="Rating v/s Number of movies (Plotly)"
)
fig.show()

```



Numpy assignments

1. Using numpy create a matrix of size 3 by 5. Create another matrix of 5 by 3

```
a= np.arange(15).reshape(3,5)
print(a)
print("data type",a.dtype)
print("Dimensions: ",a.ndim)
print("shape :",a.shape)
```

```
b= np.arange(15).reshape(5,3)
print(b)
print("data type",b.dtype)
print("Dimensions: ",b.ndim)
print("shape :",b.shape)
```

Perform following operations on the matrices

1. Display shape of both matrix

```
print("shape :",a.shape)
print("shape :",b.shape)
```

2. Find matrix multiplication

```
print(a.dot(b))
```

3. Create another matrix of size 3 by 5 using random numbers

```
c = np.random.randint(1, 100, size=(3, 5))
print(c)
```

```
[[73 52  2 48 92]
 [79 45 37 36 60]
 [15 31 42 90  5]]
```

4. And display addition subtraction and member wise multiplication

```
d=a+c
e=d-c
f=a*c
```

```
print(d)
print(e)
print(f)
```

2. accept 5 numbers from user and store it in a list1
accept 5 numbers from user and store it in list2
accept 5 numbers from user and store it in a list3
accept 5 numbers from user and store it in list4
convert these list into ndarray
combine list1 and list 2 in 1D array, arr1
combine list3 and list 4 in 1D array
combine these 4 arrays into 2D arrays (use stack functions)

```

def returnlist(n):
    lst = []
    for i in range(n):
        x = int(input("Enter the number"))
        lst.append(x)
    return lst

```

```

list1 = returnlist(5)
list2 = returnlist(5)
list3 = returnlist(5)
list4 = returnlist(5)

```

```

nlst1 = np.array(list1)
nlst2 = np.array(list2)
nlst3 = np.array(list3)
nlst4 = np.array(list4)

```

```

print(nlst1)
print(type(nlst1))

```

```

comb_12 = np.hstack((nlst1,nlst2))
print(comb_12)
comb_34 = np.hstack((nlst3,nlst4))
print(comb_34)

```

```

comb_1234 = np.vstack((nlst1,nlst2,nlst3,nlst4))
print(comb_1234)

```

3. Store yearwise sales of 2 dealers for tv, freeze, mobiles, washing machines, AC and dishwashers for years (2001 to 2010)

DEALER1

	tv	freeze	mobiles	washing machines	AC	dishwashers
(2001) 0	10000	2000	30000	4000	2000	3000
(2002) 1	10000	2000	30000	4000	2000	3000

```

import pandas as pd
import numpy as np

```

```

years = list(range(2001, 2011))
products = ['tv', 'freeze', 'mobiles', 'washing_machines', 'AC', 'dishwashers']

```

```
df_d1 = pd.DataFrame(np.random.randint(1000, 50000, size=(len(years), len(products))),
index=years, columns=products)
```

```
df_d2 = pd.DataFrame(np.random.randint(1000, 50000, size=(len(years), len(products))),
index=years, columns=products)
```

Answer the following

a. Find yearwise sales of tv,freeze,mobile,..... product for both dealer.

```
print("Products:", products)
print("Years:", years)

print("\n--- Dealer 1 Yearwise Sales (NumPy Array) ---")
print(np_d1)

print("\n--- Dealer 2 Yearwise Sales (NumPy Array) ---")
print(np_d2)
```

b. Find in each year how many products are sold by dealer1 and how many products are sold by dealer 2.

```
d1_yearly_total = np_d1.sum(axis=1)
d2_yearly_total = np_d2.sum(axis=1)
print(d1_yearly_total)
print(d2_yearly_total)

print("\n--- Total Products Sold per Year ---")
for i, y in enumerate(years):
    print(
        f"Year {y} -> Dealer 1: {d1_yearly_total[i]} | Dealer 2: {d2_yearly_total[i]}"
    )
```

c. Display average sales of a 2005 year for each dealer.

```
d1_yearly_Avg = np_d1.mean(axis=1).round(2)
d2_yearly_Avg = np_d2.mean(axis=1).round(2)
print(d1_yearly_Avg)
print(d2_yearly_Avg)

print(
    f"Year Avg {2005} -> Dealer 1: {d1_yearly_Avg[4]} | Dealer 2: {d2_yearly_Avg[4]}"
)
```

d. Display max sales of a 2005 year for each dealer

```

d1_yearly_Max = np_d1.max(axis=1).round(2)
d2_yearly_Max = np_d2.mean(axis=1).round(2)
print(d1_yearly_Max)
print(d2_yearly_Max)

print(
    f"Year Max {2005} -> Dealer 1: {d1_yearly_Max[4]} | Dealer 2: {d2_yearly_Max[4]}"
)

```

e. Display sales of TV and fridge for each dealer.

```

d1_tv_fridge = np_d1[:,2]
d2_tv_fridge = np_d2[:,2]

print("\n--- Dealer 1: TV & Fridge Sales (2001-2010) ---")
print("Year | TV | Fridge")
for i, y in enumerate(years):
    print(f"{y} | {d1_tv_fridge[i, 0]} | {d1_tv_fridge[i, 1]}")

print("\n--- Dealer 2: TV & Fridge Sales (2001-2010) ---")
print("Year | TV | Fridge")
for i, y in enumerate(years):
    print(f"{y} | {d2_tv_fridge[i, 0]} | {d2_tv_fridge[i, 1]}")

```

f. Draw 2 bar chart one for each dealer, for each year total number of items sold in the year

```

import matplotlib.pyplot as plt

d1_yearly_total = np_d1.sum(axis=1)
d2_yearly_total = np_d2.sum(axis=1)

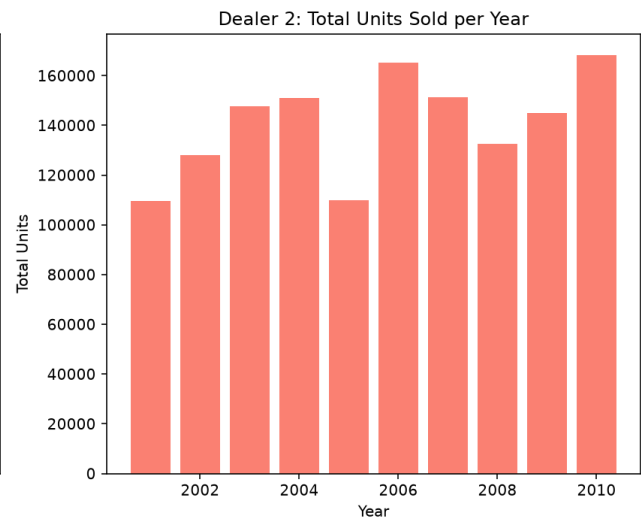
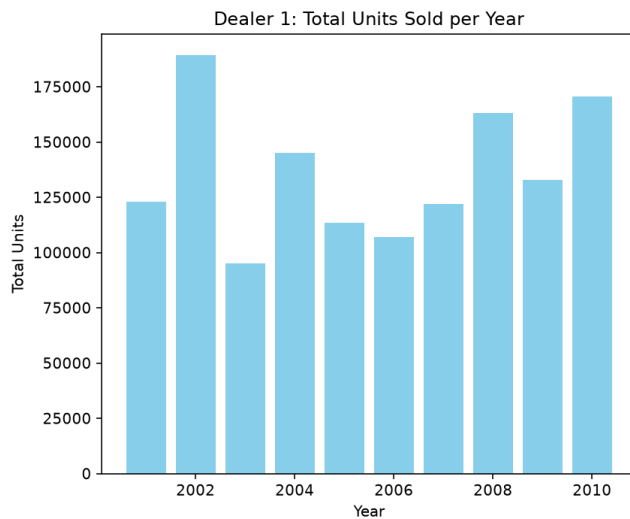
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.bar(years, d1_yearly_total, color="skyblue")
plt.title("Dealer 1: Total Units Sold per Year")
plt.xlabel("Year")
plt.ylabel("Total Units")

plt.subplot(1, 2, 2)
plt.bar(years, d2_yearly_total, color="salmon")
plt.title("Dealer 2: Total Units Sold per Year")
plt.xlabel("Year")
plt.ylabel("Total Units")

plt.tight_layout()
plt.show()

```



g. Draw a pie chart one for each dealer, total number of items sold in the year

```
d1_yearly_total = np_d1.sum(axis=1)
```

```
d2_yearly_total = np_d2.sum(axis=1)
```

```
plt.figure(figsize=(12, 6))
```

```
plt.subplot(1, 2, 1)
```

```
plt.pie(d1_yearly_total, labels=years, autopct="%1.1f%%", startangle=140)
```

```
plt.title("Dealer 1: Product Sales Share (2001-2010)")
```

```
plt.subplot(1, 2, 2)
```

```
plt.pie(d2_yearly_total, labels=years, autopct="%1.1f%%", startangle=140)
```

```
plt.title("Dealer 2: Product Sales Share (2001-2010)")
```

```
plt.tight_layout()
```

```
plt.show()
```

